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Project Title:	Acute and Long Term Outcomes of Simulated Deep Space Radiation Exposure on Latent Viral CNS Infection and CNS Pathology		
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COI Name (Institution):	Thompson, Richard Ph.D. (University of Cincinnati) Williams, Michael Ph.D. (Children's Hospital Medical Center, Cincinnati)		
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Task Description:	The successful diaspora of humans into deep space requires an understanding of the risk of deep space radiation exposure on short and long term health outcomes, including risks to the central nervous system (CNS). Assessing CNS risk must consider the impact of deep space radiation on the behavior of latent neurotropic infectious agents that reside within the human CNS. Herpes simplex virus (HSV) is among the most widespread agent in this category, permanently infecting the nervous systems of more than 80% of the global population. Latent HSV cannot be eliminated from the nervous system and there is no vaccine to prevent infection. Thus for the foreseeable future, humans traveling into deep space will harbor this and other alpha-herpesviruses. While uncontrolled HSV replication in the CNS results in often fatal encephalitis, this virus also has the capacity to promote chronic inflammation and neurodegeneration. Indeed there is substantial evidence that in combination with genetic risk factors, HSV infection in the CNS further increases the probability of developing Alzheimer's disease. In this proposal, the effects of simulated deep space radiation on the development of CNS pathology (reactive lesions) and the impact of latent and induced in vivo reactivation of HSV infection on the development of CNS pathology will be determined together with the role that human ApoE alleles play in these processes. We will employ our well characterized mouse model of HSV latency and in vivo reactivation in combination with quantitative measures of reactive lesions in the CNS to determine the effect of simulated deep space radiation on HSV mediated CNS damage. The acute stage of infection will be monitored using our standard approaches including quantitative evaluation of infectious virus titers. Latent infection in the trigeminal ganglia and brain will be quantified at 30-40 days postinfection (pi) using our quantitative real time PCR assay for the viral genome. Spontaneous and induced exit from latency will be quantified at the single neuron level using our whole TG immunohistochemical assay. This approach permits us to identify even a single positive neuron in groups of ganglia. Viral reactivation (production of infectious virus) will be evaluated using our standard assay. Reactivation will be induced using our well-established hyperthermic stress procedure (temperature of the mice is raised to 42.5 C for 10 minutes). We have shown and published that this procedure reliably induces viral reactivation and can be repeatedly employed two times per week for periods of up to at least one year pi. We have determined the repeated reactivation stress of C57Bl/6 mice results in reactive astrocytic lesions in the CNS of HSV infected but not in uninfected mice. Interestingly this damage is much more extensive in mice that express the human ApoE4 allele or in IL10 knock out mice. This suggests important relevance to human biology since both IL10 polymorphisms and the ApoE4 allele have been shown to be associated with increased risk of acquiring Alzheimer's disease. We now propose to collaborate with radiation biologists at NASA to determine appropriate conditions to simulate deep space radiation and we will quantify the effects of such radiation on CNS damage induced by reactivation of HSV latent infections. Significance: The goal of NASA's space radiation research is to enable the human exploration of space within acceptable risks from space radiation. Importantly, the proposed studies will determine the integrated risk of the interaction of galactic cosmic radiation with uninfected and latently infected CNS. The prevalence of HSV infection among humans makes this a nearly unavoidable additional factor that could significantly heighten the risk of CNS damage upon GCR (galactic cosmic radiation) exposure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Importantly, beyond its significance to deep space exploration, to our knowledge, this project is the first to systematically examine the long term impact of HSV latent/reactivation cycles in the nervous system in an experimental model system and test the additional risk of carrying the huAPOE4 allele. These studies will therefore pioneer a model that will be extremely useful in understanding the long term outcomes of latent HSV in the nervous system among humans on Earth. Understanding the consequences on long term latency/reactivation is the first step toward developing treatment and preventive strategies. We have found both memory impairment and striking Alzheimer's Disease associated changes in the CNS of the HSV long term infected but not the mock infected CNS.
	FINAL REPORTING SEPTEMBER 2018: Rationale: Comprehensive risk assessment of deep space radiation exposure on the central nervous system (CNS) must consider the high probability of the presence of latent (persistent) infectious agents that reside within the human CNS and for which there are no cures. Herpes simplex virus (HSV) is among the most widespread agents in this category with >80% of the global human population above the age of 20 infected with this virus. Infection includes efficient invasion of the nervous system and deposition of millions of viral genomes in many 1,000s of neurons. These genomes serve as a life-long reservoir from which the virus can enter the viral lytic cycle producing virus which can be transmitted to new hosts. Reactivation of latent HSV (the frequency of that is stress related) results in chronic inflammation in the peripheral nervous system, in the eye where it can cause herpetic stromal keratitis, and potentially in the brain. Infections that remain benign for decades can result in uncontrolled viral replication and serious life threatening outcomes, including encephalitis for which the precipitating factors are often not known. Goal of the project: The goal of this project was to assess the risk of combined simulated galactic cosmic radiation (sGCR) exposure and the presence of latent HSV in the nervous system. Using a well-characterized model of acute and latent HSV infection in the mouse as our working platform, risk associated with sGCR exposure in the context of the full complexity of host responses in the presence and absence of viral latency was evaluated. Two time frames were investigated: (i) acute effects, i.e., does sGCR cause viral reactivation in the latently infected nervous system, and if so, what is the outcome, and (ii) long term effects, i.e., are long term outcomes of latent virus in the nervous system altered by sGCR exposure. Four major findings have come from our studies: 1. sGCR exposure of HSV latently infected mice resulted in no widespread lytic activation of latent viral genomes in the nervous system but did result in the highly restricted and temporally limited reactivation associated with terrestrial stimuli (i.e., occurring in 1-3 neurons and resolving in the nervous system within 48 hrs). 2. Our studies modelling moderate immune suppression (a known space mission risk) combined with HSV reactivation (space mission risk) revealed that moderate immune suppression does not alter the control of HSV spread following a reactivation event. In contrast, this same immune suppression resulted in severe disease outcomes in newly HSV infected hosts. Thus HSV latent infection and commensurate immunity to HSV, even considering the potential for reactivation from latency, may be an advantage to crew unless all crew are uninfected or a vaccine becomes available. 3. Despite reduced spleen weight in sGCR exposed group, immunity to HSV was fully maintained. Thus reduced spleen

Task Progress:	weight did not correlate with a detectable defect in long term maintenance of immunologic memory or the downstream ability to mount a protective immune response. 4. Long term HSV latency results in cognitive impairment, a finding observed in 3 independent groups in both sGCR exposed and unexposed groups. However, these behavioral effects were not significantly exacerbated by radiation exposure. The cognitive impairment was associated with AD like changes in the brain and especially prominent in the huApoE4 targeted replacement mice. This model, combining two known AD risk factors, aging and the ApoE4 allele with a latent viral infection is an important advance. Radiation exposure had some effect independent of infection, but with rare exceptions, did not significantly exacerbate the effects of infection alone. ANNUAL REPORTING OCTOBER 2017: Comprehensive risk assessment of deep space radiation exposure on the central nervous system (CNS) must consider the high probability of the presence of latent (persistent) infectious agents that reside within the human CNS and for which there are no cures. Herpes simplex virus (HSV) is among the most widespread agents in this category with >80% of the global human population above the age of 20 infected with this virus. Infection includes efficient invasion of the nervous system and deposition of millions of viral genomes in 10s of 1,000s of neurons. These genomes serve as a life-long reservoir from which the virus can enter the viral lytic cycle producing virus which can be transmitted to new hosts. Reactivation of latent HSV (the frequency of which is stress related) results in chronic inflammation in the peripheral nervous system, in the eye where it can cause herpetic stromal keratitis, and in the brain. However, infections that remain benign for decades can result in uncontrolled viral replication and serious life threatening outcomes, including encephalitis. The goal of this project is to assess the risk of combined simulated galactic cosmic radiation (sGCR) exposure and the presence of latent HSV in the nervous system. Two time frames have been investigated: (i) acute effects, i.e., does sGCR cause viral reactivation in the latently infected nervous system, and if so, what is the outcome, and (ii) long term effects, i.e., are long term outcomes of latent/lytic cycles in CNS altered by sGCR exposure. Using a well-defined experimental mouse model of HSV latency/reactivation we have identified the following outcomes. • In two independent studies, sequential exposure to 60cGy H+ and 20cGy 56Fe of either mock or HSV latently infected mice revealed no adverse acute health outcomes (0-3 weeks post irradiation). However, we did find that H+ exposure perturbs viral latency resulting in transition into the viral lytic cycle 22-24 hrs post exposure raising an additional concern. Although proton induced reactivation appeared similar in nature as reactivation following “terrestrial” in vivo reactivation stimuli (very low number of neurons and strict containment of viral spread within the ganglion), an important question is the risk of this H+ induced reactivation event in the context of the well-documented mission related immune suppression. We approached this experimentally, simulating three relevant scenarios to evaluate risk associated with latent HSV infection combined with immune-suppression. These scenarios represent immune suppression at the time of (i) a reactivation event, (ii) re-exposure (modeling potential transmission of virus shed from crew mate), and (iii) acute exposure (modeling transmission of virus shed from crew mate to naïve crew member). Approach: Three strategies for generating immune-suppression of mice were utilized including T cell depletion (CD4, CD8, and CD45RB subsets), cyclophosphamide treatment, and specific interferon depletion. Outcome: Although the efficacy of treatment on central immune cell depletion was documented, in the context of either induced reactivation or re-exposure, reactivation in the nervous system was controlled normally suggesting that tissue resident T cells are the critical players. In contrast, immune-suppression of HSV naïve mice followed by infection resulted in severe disease outcomes. These findings suggest that the immunity engendered by HSV infection may be sufficiently robust to afford the crew protection from reinfection by virus shed from other crew members and prevent uncontrolled spread of reactivated virus in the nervous system despite a level of mission related immune suppression, recognizing that our immune suppression may not fully parallel that which occurs in space. • A significant reduction in spleen weight at 12 months in irradiated compared with nonirradiated mice was observed. The risk associated with this effect was evaluated using a sensitive HSV reinfection assay. This assay demonstrated that no defect in the ability to mount a potent protective immune response was observed in these radiation-exposed mice. Thus reduced spleen weight did not correlate with a defect in long term maintenance of immunologic memory or the downstream ability to mount a protective immune response. • From 4-16 months, with the exception of eye disease, severe outcomes (mortality or severe morbidity) were not prevalent in any group. Nevertheless, in both experiments, long term HSV latent infection correlated with significant behavioral deficits. These deficits were exacerbated in the background of the huApoE4 targeted replacement (TR) mouse, further supporting the link between HSV latent infection and this allele with an increased risk of neurodegenerative disease. However, at the level of this assessment, our findings did not support the hypothesis that radiation exposure significantly exacerbated behavioral effects. • Evaluation of CNS revealed increased gliosis and presence of additional markers of neurodegeneration (also age related). One notable observation is a striking increase in intracellular iron in neurons within the trigeminal ganglia and the CNS of HSV latently infected mice. Importantly, sGCR exposure resulted in a dramatic shift in the intracellular localization of iron from the nucleus into the cytoplasm. Considering the potential link between iron and neurodegenerative disease as well as the established microgravity driven iron tissue overload during long term space flight, these observations warrant follow-up to understand the risk of combined latent HSV infection and travel into deep space. Our data support the conclusion that despite the absence of signs of HSV reactivation (surface lesions or viral shedding), there is evidence for a destructive process associated with latent infection in the CNS. To our knowledge, these are the first systematic studies addressing the behavioral/cognitive implications of long term latent HSV infection. Understanding this process, the factors influencing susceptibility and strategies to ameliorate this process are important goals for the population at large as well as mission crew.
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